

E-Commerce Dashboard Explainer

Pages 1 to 4: Executive Summary, Drivers, Cohorts, Customer Experience

Generated: February 09, 2026

This document describes the purpose and interpretation of each dashboard element across Pages 1 to 4. It is written to be easy to paste into presentation notes and to help reviewers understand what each visual is saying and why it exists.

Data model note (high level): Page KPIs are powered by curated mart views in ecom_mart. Key facts include an item grain order item fact (with derived delivery and lateness fields) and an order grain fact covering the full order universe.

Terms used in this dashboard

Term	Meaning
AOV	Average Order Value; here typically computed on orders with items (revenue / orders).
Delivered rate	Share of orders or items marked delivered (delivered timestamp present).
Late rate among delivered	Among delivered, share delivered after estimated delivery date.
Cohort month	Month of a subscriber's first observed order.
Month_n	Month offset since first purchase month (Month +1 through Month +12).

Page 1: Executive Summary and Trends

This page answers: How is the business performing overall, and how are the core KPIs trending? It is designed to give a fast read before drilling into drivers or cohorts.

KPI Counter Grid (2 by 4)

What it shows: A compact set of headline KPIs: total orders, total item revenue, AOV (orders with items), delivered rate, late rate among delivered, payment rate, review rate, and average review score over reviewed orders.

How it's used: A first glance health check. Reviewers can scan for sudden shifts (for example, revenue up but delivery reliability down) before looking at trend charts.

Primary source: ecom_mart.v_kpi_overview

Revenue Sparkline (Last 12 active months)

What it shows: A short revenue trend for months with real activity (filters out tail ghost months where metrics are zero or null).

How it's used: A quick view of momentum and seasonality without being distracted by incomplete data months; also validates that the trend window is stable.

Primary source: ecom_mart.v_kpi_monthly (active month filter)

Monthly Trend (Revenue, Orders, AOV)

What it shows: Time series showing how revenue, orders, and AOV move month to month (active months only).

How it's used: Separates volume vs basket effects: AOV rising with flat orders suggests higher order value; rising orders with flat AOV suggests volume driven growth.

Primary source: ecom_mart.v_kpi_monthly

New vs Returning Revenue Composition

What it shows: Monthly revenue split between new and returning customers, where customer type is based on stable subscriber identity (not per order transaction id).

How it's used: Separates acquisition driven revenue from repeat driven revenue; helps detect whether growth depends on constantly new customers (riskier) vs returning behavior (stickier).

Primary source: ecom_mart.v_kpi_new_vs_returning_revenue_monthly (subscriber based)

Delivery Health (Split charts)

What it shows: Delivered rate and late rate among delivered shown on separate lines so the small late rate signal remains readable next to a high delivered rate baseline.

How it's used: Monitors customer experience: delivered rate answers did it arrive at all; late rate answers did it arrive on time. Tooltips may include average delivery days for context.

Primary source: ecom_mart.v_kpi_delivery_monthly

Page 2: Drivers (Month filtered)

This page answers: What is driving performance in the selected month? A month filter applies only to Page 2 so the analysis stays apples to apples within the chosen month.

Top Categories (Top 10 table)

What it shows: Ranks product categories by revenue for the selected month, with supporting metrics like orders and average review score. Category names are cleaned for readability.

How it's used: Identifies where revenue is concentrated and whether top categories are also high quality (reviews) or risky. Revenue per order helps compare categories fairly.

Primary source: `ecom_mart.v_top_categories_monthly`

Top Sellers (Top 10 table)

What it shows: Ranks sellers by revenue for the selected month, with orders, items, average review score, and late rate among delivered.

How it's used: Surfaces key sellers powering the month and operational risk (high late rate) among high volume sellers; useful for partner management and performance monitoring.

Primary source: `ecom_mart.v_top_sellers_monthly`

Payment Mix (Snapshot for selected month)

What it shows: Distribution of orders and payment value by payment type for the selected month. Payment types are cleaned (underscores removed; `not_defined` becomes `Undefined`).

How it's used: Explains checkout behavior and operational dependencies; also serves as a data quality check if `Undefined` spikes.

Primary source: `ecom_mart.v_kpi_payment_mix_monthly` (active month aligned)

Geo Revenue (Country level)

What it shows: Revenue, orders, and supporting metrics by customer geography, kept at the country level for clarity.

How it's used: Detects geographic concentration and top markets; can guide shipping and marketing focus and where delivery performance should be scrutinized.

Primary source: `ecom_mart.v_kpi_geo_revenue_monthly`

Revenue Concentration (Pareto curve)

What it shows: Cumulative revenue share by subscriber rank (highest revenue subscribers first).

Observed concentration: top 10 percent is about 38 percent of revenue; top 20 percent is about 53.66

percent.

How it's used: Quantifies customer concentration risk and whether revenue depends heavily on a small set of subscribers. This tile is intentionally not tied to the month filter to reflect overall concentration.

Primary source: ecom_mart.v_kpi_revenue_concentration_subscribers

Page 3: Cohorts

This page answers: How do customers behave over time after their first purchase? Cohort analysis groups subscribers by their first purchase month and tracks retention and value in subsequent months.

Retention Heatmap (Months +1 to +12)

What it shows: For each cohort (row), shows the share of subscribers who place an order in each month offset after first purchase (columns). Month +0 is excluded because it is always 100 percent.

How it's used: Compares stickiness across cohorts. Darker cells indicate more repeat purchasing at that offset. Recent cohorts naturally have fewer observed months, creating shorter rows unless the grid is explicitly filled.

Primary source: `ecom_mart.v_kpi_subscriber_cohort_retention_12m`

Cohort Revenue Heatmap (Months +1 to +12)

What it shows: Total revenue generated by each cohort in each month offset after first purchase.

How it's used: Shows which cohorts keep generating dollars over time (a blend of cohort size and repeat behavior); helps spot strong acquisition months that continue to monetize.

Primary source: `ecom_mart.v_kpi_subscriber_cohort_revenue_12m`

Cohort Size Trend

What it shows: Number of subscribers entering each cohort month (first purchase month).

How it's used: Adds context to retention and revenue patterns. Large cohorts create more reliable trends; tiny cohorts can show extreme rates due to small denominators.

Primary source: `ecom_mart.v_kpi_subscriber_cohort_sizes`

Delivery Speed by Order Approval Month

What it shows: Average delivery time (days) for delivered items, grouped by the month the order was approved.

How it's used: Monitors operational performance over time and ties experience shifts to seasonality or logistics changes; good early warning signal for satisfaction issues.

Primary source: Query from `ecom_mart.fct_order_items_final` (delivered only)

Late Rate Among Delivered Orders (Monthly)

What it shows: Percent of delivered orders that arrive after the estimated delivery date, by month. Ghost months with zero delivered orders are filtered out.

How it's used: Reliability metric distinct from delivered rate; helps explain churn or review drops and supports seller and logistics investigations.

Primary source: ecom_mart.v_kpi_delivery_monthly (filtered to orders_delivered > 0)

Payment Method Mix Over Time (100 percent stacked)

What it shows: Monthly share of orders by payment method, with cleaned labels (for example, Credit Card, Undefined). Some early months may appear blank if there are no recorded orders or payments.

How it's used: Detects shifts in checkout behavior, promo effects (vouchers), or data quality issues (Undefined rising). Complements Page 2's month snapshot with longer term movement.

Primary source: ecom_mart.v_kpi_payment_mix_monthly

Page 4: Customer Experience Deep Dive

This page answers: How does the delivery experience feel for customers, and which sellers or regions most contribute to late delivery and weaker satisfaction signals? Unlike Page 2 (Drivers), these elements are designed to be stable and interpretable without relying on a month filter.

Delivery Speed Percentiles (KPI cards)

What it shows: P50 (median), P90, and P95 delivery days for delivered items. P50 reflects the typical customer experience; P90 and P95 reflect the tail (slow deliveries that a meaningful share of customers still encounter).

How it's used: Summarizes speed and variability without being distorted by extreme outliers. A large gap between P50 and P90 or P95 indicates a long tail of slow deliveries.

Primary source: Query from `ecom_mart.fct_order_items_final` (delivered only; `delivery_days` not null)

Late Rate (Delivered Only) (KPI card)

What it shows: Percent of delivered items that arrived after the estimated delivery date.

How it's used: Headline reliability metric (did it arrive on time). Pairs with the percentiles: percentiles describe speed; late rate describes promise adherence.

Primary source: Query from `ecom_mart.fct_order_items_final` (delivered only)

At Risk Sellers (Late Delivery Impact) (Top 50)

What it shows: Ranked list of the sellers most likely driving late deliveries at scale. Late Rate is the seller's late rate among delivered. Impact is a volume weighted proxy ($\text{Late Rate} \times \text{Orders}$) used for prioritization.

How it's used: Focuses operational follow up where it matters most: high late rate combined with meaningful order volume. A minimum volume threshold reduces small sample noise.

Primary source: `ecom_mart.v_top_sellers_monthly` (rolled up across active months) plus active month filter from `ecom_mart.v_kpi_monthly`

Review Score vs Lateness (Seller Scatter)

What it shows: Each point is a seller. X axis is late rate among delivered; Y axis is average review score (weighted by orders). Point size can be set to orders to reflect impact.

How it's used: Identifies customer experience risk sellers (high late rate and lower review scores) and helps assess whether lateness appears associated with satisfaction outcomes.

Primary source: `ecom_mart.v_top_sellers_monthly` (rolled up across active months) plus active month filter from `ecom_mart.v_kpi_monthly`

Late Rate by Seller Country

What it shows: Late rate among delivered aggregated by seller_country (orders weighted), with volume context (orders and revenue).

How it's used: Detects whether late delivery risk is concentrated in specific seller regions, supporting targeted logistics or onboarding improvements.

Primary source: ecom_mart.v_top_sellers_monthly (rolled up across active months) plus active month filter from ecom_mart.v_kpi_monthly